

the histone core. It is important to keep in mind that these enzymes can be involved both in gene-activation events, by facilitating the binding of transcriptional activators, and in gene-repression events, perhaps by facilitating the binding of a transcriptional repressor or by directly promoting compaction of the chromatin structure. SEE ALSO CELL CYCLE; CELL, EUKARYOTIC; CHROMOSOMAL BANDING; CHROMOSOME, PROKARYOTIC; DNA; EVOLUTION OF GENES; GENE; GENE EXPRESSION: OVERVIEW OF CONTROL; *IN SITU* HYBRIDIZATION; MEIOSIS; MITOSIS; MOSAICISM; REPETITIVE DNA ELEMENTS; REPLICATION; TELOMERE; X CHROMOSOME; Y CHROMOSOME.

Cynthia Guidi and Anthony N. Imbalzano

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Chromosome, Prokaryotic

The bacterial or prokaryotic chromosome differs in many ways from that of the eukaryote. The term "eukaryote" comes from the Greek and means "true nucleus." Eukaryotic cells have a double membrane (the **nuclear membrane**) surrounding the nucleus, the organelle that contains several chromosomes. In contrast, the term "prokaryote" means "primitive nucleus," and, indeed, cells in prokaryotes have no nucleus. Instead, the prokaryotic chromosome is dispersed within the cell and is not enclosed by a separate membrane.

This dispersed chromosome is called the bacterial "nucleoid," which can be seen in electron micrographs of thin sections, as shown in Figure 2. Although bacteria (now called eubacteria) are highly diverse, the prototypical bacterial species is *Escherichia coli*, which has served as a model organism for genetic, biochemical, and biotechnological research for many decades.

The *E. coli* chromosome is a single circle. Because the single DNA molecule forming the chromosome is so long (about 4.6 million base pairs), it is easily broken when researchers try to isolate it. However, in the early 1960s, the Australian biochemist John Cairns was able to gently **lyse** *E. coli* cells without breaking the chromosome. He was interested in chromosomal replication and had labeled the DNA with tritium (^3H), a radioactive form of hydrogen. Autoradiograms of the DNA demonstrated that the bacterial chromosome is a circular molecule. While the vast majority of bacterial species possess a single unique chromosome, there are a few rare species,

nuclear membrane
membrane surrounding
the nucleus

lyse break apart



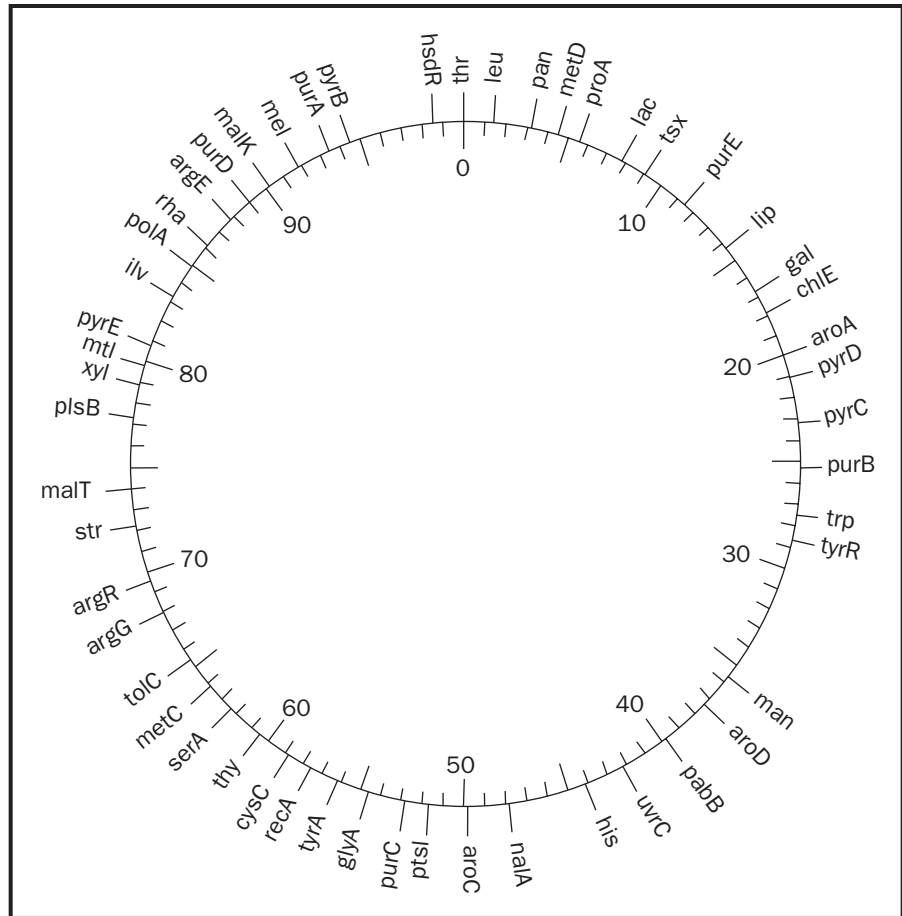


Figure 1. The circular genetic map of the *Escherichia coli* chromosome. The map is divided into 100 map units, and representative genes are shown. The *lac* operon is at approximately 8 minutes. The origin of replication (not shown) is at 83.5 minutes. Based on Ingraham, 2000.

such as *Vibrio cholerae* (the agent that causes the disease cholera) and *Deinococcus radiodurans*, that have two different chromosomes.

It is also quite common for bacterial species to possess extrachromosomal genetic elements called plasmids. These are small, circular DNA molecules which, when present, vary in number from one to about thirty identical copies per cell. Plasmids include the fertility factor (F^+ plasmid), described below, as well as plasmids that carry drug-resistance genes. Indeed, these drug-resistance plasmids may be passed from species to species and are a major problem in the spread of antibiotic resistance. Whereas most bacteria that contain plasmids have just a single kind of plasmid, some bacterial species simultaneously possess a number of different plasmids, each of which, in turn, is present in varying numbers within the bacterial cell.

The bacterial chromosome is condensed into chromosomal domains.

The bacterial chromosome must be tightly packed to fit into the small volume of the bacterial cell. Figure 3 shows the relative sizes of the unfolded chromosome and the *E. coli* cell. During the 1980s, techniques were developed to isolate intact bacterial nucleoids by gentle lysis, under conditions that prevented the DNA of the chromosome from uncoiling. These isolated



Figure 2. A transmission electron microscope image of *Escherichia coli* magnified approximately 20,000 times.

nucleoids were highly condensed into a very compact structure, as shown in Figure 4.

Compacting the DNA involves supercoiling, or further twisting the twisted chromosome. The chromosome's fifty or so DNA domains are held together by a scaffold of RNA and protein, and the entire nucleoid is attached to the cell membrane. This membrane attachment aids in the segregation of the chromosomes after they replicate in preparation for cell division. Bacteria lack the histone proteins that are found bound to the DNA and that form the nucleosomes of eukaryotic chromosomes. However, it is believed that polyamines (organic molecules with multiple $-NH_2$ amine groups) such as spermidine, as well as some basic proteins, aid in compacting the bacterial chromosome. These basic proteins have a net positive charge that bind them to the negative charge of the phosphates in the DNA backbone.

Replication of the circular chromosome begins at a single point, called OriC, and proceeds in both directions around the circle, until the two replication forks meet up. The result is two identical loops. Replication takes approximately forty minutes.

The *E. coli* genetic chromosome. The field of bacterial genetics began in 1946, even before the structure of DNA was determined, with the discovery by the geneticists Joshua Lederberg and Edward Tatum at the University of Wisconsin of sex in bacteria, in the form of conjugational genetic exchange between *E. coli* bacteria. In the conjugation process, a fertility



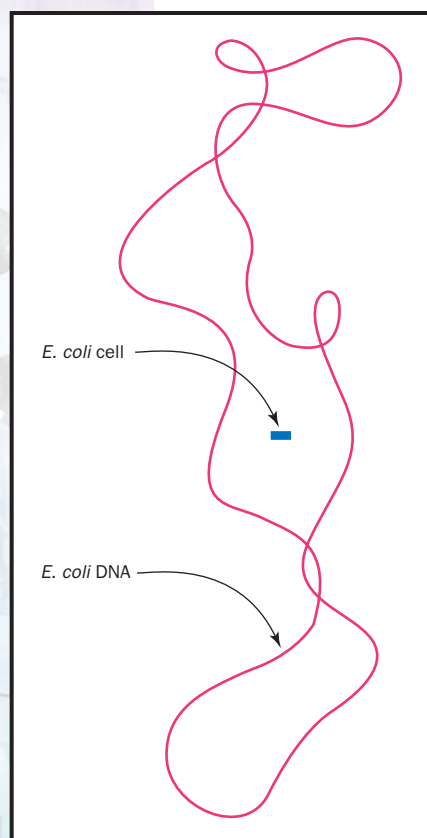


Figure 3. Relative size of an *E. coli* cell and its chromosome.

pathogens disease-causing organisms

factor (F plasmid) recombines with (splices itself into) the *E. coli* chromosome at a specific site. It then acts as a “molecular motor” to drive the transfer of the entire *E. coli* chromosome to a recipient (F^-) cell. The transferred molecule can then recombine with the host chromosome, increasing the genetic diversity of the host. Transferring the entire chromosome takes approximately one hundred minutes, and thus the genetic map is divided into one hundred minutes (which were later defined as one hundred map units). As more and more genetic markers were found and mapped, it became apparent that the genetic chromosome map formed a circle, as shown in Figure 1.

The DNA sequence of the *E. coli* chromosome. *E. coli* was chosen as one of the genetic model organisms whose chromosome was to be sequenced as part of the Human Genome Project. Although it was not the first bacterial species to be completely sequenced, it was one of the most important ones. In 1997, Fredrick Blattner of the University of Wisconsin and colleagues published the sequence of 4,639,221 base pairs of the K-12 laboratory strain. *E. coli* is estimated to have 4,279 genes.

Many sets of genes on the *E. coli* chromosome are organized into operons. An operon is a set of functionally related genes that are controlled by a single promoter and that are all transcribed at the same time.

Comparative bacterial genomes. As of June 2002, the genomes of sixty-five different bacterial species had been completely sequenced. Several of these are listed in Table 1, along with the genomes’ size and number of genes. Many of the species sequenced are human **pathogens**. Having the DNA sequence will prove useful in designing drugs and antibiotics to combat infections and bacterial toxins. DNA sequences may be found on the Internet, at the Genome Web site of the National Center for Biotechnology Information.

CHROMOSOME SIZE AND NUMBER OF GENES FOR SEVERAL BACTERIAL SPECIES SEQUENCED 1995–2000

Bacterial species	Chromosome size (base pairs)	Number of genes	Year sequence completed
<i>Haemophilus influenzae</i>	1,830,138	1714	1995
<i>Mycoplasma genitalium</i>	580,074	480	1995
<i>Synechocystis</i> sp.	3,573,470	3167	1996
<i>Mycoplasma pneumoniae</i>	816,394	1054	1996
<i>Helicobacter pylori</i>	1,667,867	1576	1997
<i>Escherichia coli</i>	4,639,211	4279	1997
<i>Bacillus subtilis</i>	4,214,814	4112	1997
<i>Borrelia burgdorferi</i>	910,724	851	1997
<i>Aquifex aeolicus</i>	1,551,335	1529	1998
<i>Mycobacterium tuberculosis</i>	4,411,529	3927	1998
<i>Treponema pallidum</i>	1,138,011	1036	1998
<i>Rickettsia prowazekii</i>	1,111,523	835	1998
<i>Chlamydia trachomatis</i>	1,042,519	895	1998
<i>Chlamydia pneumoniae</i>	1,230,230	1054	1999
<i>Thermotoga maritima</i>	1,860,725	1858	1999
<i>Campylobacter jejuni</i>	1,641,481	1654	2000
<i>Neisseria meningitidis</i>	2,272,351	2079	2000
<i>Buchnera</i> sp.	640,681	564	2000
<i>Bacillus halodurans</i>	4,202,353	4066	2000

Table 1.

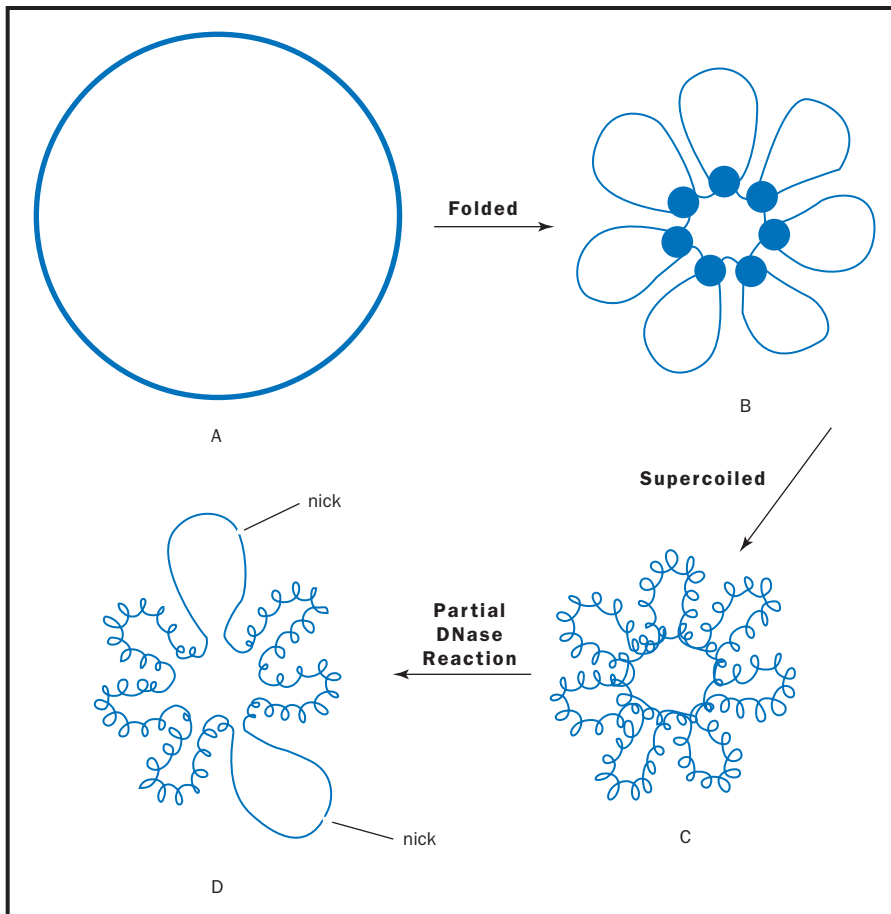


Figure 4. Model of the condensed bacterial chromosome. The circular DNA molecule (a) is folded into several domains (b). Seven domains are shown, but in the cell there are approximately fifty. These domains are held together by RNA and proteins. The DNA is further condensed by supercoiling (c). If the DNA backbone is nicked by a nuclease, the DNA domain opens (d). Adapted from Pettijohn, David E., 1996.

Minimal-gene-set concept. One of the interesting features of studying bacterial chromosomes has been the concept of the minimal number of genes a cellular life form would need to survive. (This excludes viruses and viroids, which need living cells of a host in which to carry out their life cycle.) We know from the sequence of the *Mycoplasma genitalium* chromosome, the smallest genome sequenced so far, that the upper limit of the minimal gene set is 480, as shown in Table 1. After the sequence of the *Haemophilus influenzae* chromosome was completed, a comparison of the genes that were identical (or highly conserved) in the two species led to an estimate of 256 as the minimal gene set. The National Institutes of Health scientist Eugene Koonin, with the availability of many more sequenced species, has also estimated a minimal size of about 250 genes. It may be possible in the future for scientists to construct a minimal life-form by removing nonessential genes from an organism such as *M. genitalium*. SEE ALSO ANTIBIOTIC RESISTANCE; ARCHAEA; CHROMOSOME, EUKARYOTIC; CONJUGATION; *ESCHERICHIA COLI* (*E. COLI* BACTERIUM); EUBACTERIA; HUMAN GENOME PROJECT; OPERON; REPLICATION.

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Chromosomes, Artificial

Artificial chromosomes are laboratory constructs that contain DNA sequences and that perform the critical functions of natural chromosomes. They are used to introduce and control new DNA in a cell, to study how chromosomes function, and to map genes in genomes.

Natural Chromosome Function

DNA, which constitutes the genome of a cell, is always packaged with a variety of proteins, and together these make up the chromosomes. A chromosome serves to compact the DNA and protect it from the damage, while at the same time allowing the genes it contains to be available for transcription into RNA. In addition to these functions, extra ones are necessary when the cell divides. Prior to cell division the DNA must be copied and these copies separated (segregated) and delivered to different parts of the cell, ensuring that each of the new cells receives only a single copy.

To ensure correct segregation, chromosomes have to have distinct components that are composed of specific DNA sequences and associated proteins. Bacterial chromosomes, (**plasmids**) which are circular, have a single site at which DNA replication originates, and attachment to the cell membrane results in segregation. Artificial bacterial chromosomes (BACs) mimic this using appropriate origin sequences.

In organisms with multiple linear chromosomes (eukaryotic organisms) the process is more complicated. The ends of the chromosomes must be protected from degradation and from the mechanisms that the cell uses to protect itself against broken DNA. **Telomeres**, which provide these functions, are arrays of short, repeated sequences with complexes of specific proteins attached. To ensure segregation complexes of other proteins, DNA sequences known as kinetochores form at sites known as **centromeres**. These contain molecular motors, systems to monitor correct segregation, and sites for attachment of **microtubules**. Chromosomes will contain one or more origins of replication.

Yeast Artificial Chromosomes

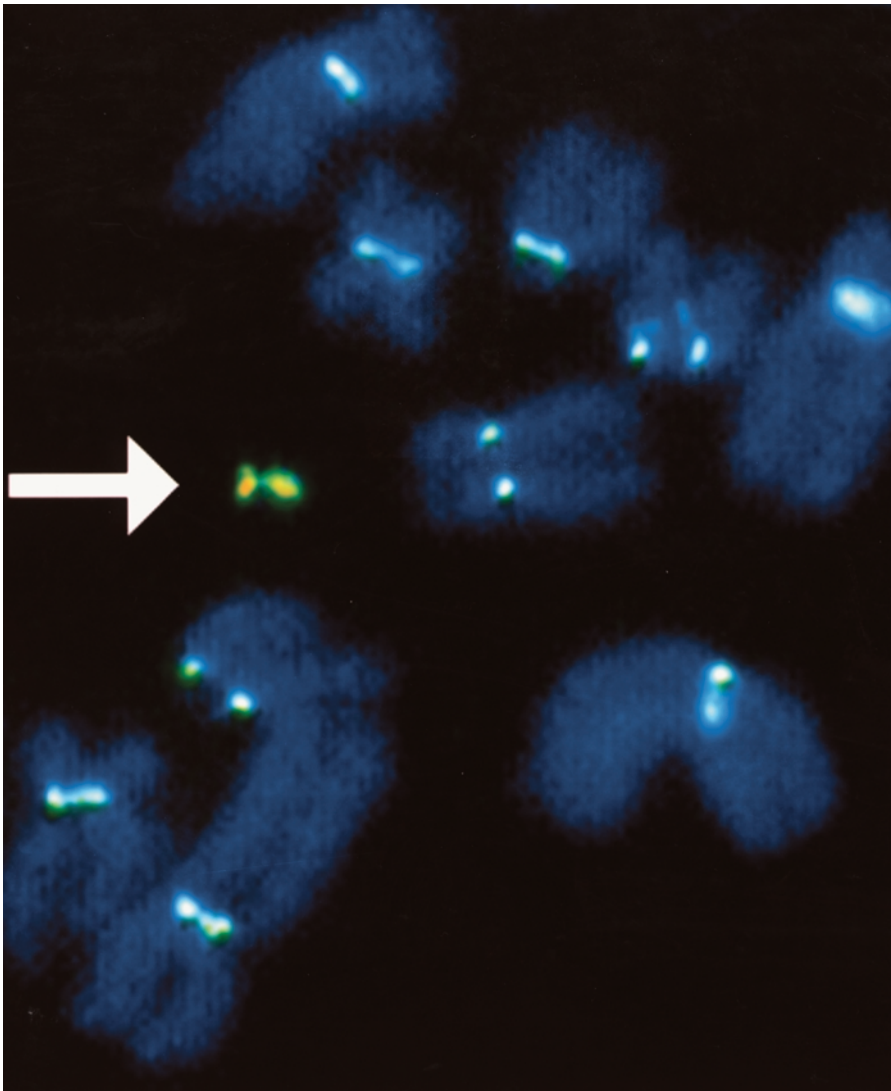
Artificial chromosomes for use in yeast and mammalian cells aim to replicate these components on a single DNA molecule. In bakers yeast (*Saccharomyces*

plasmids small rings of DNA found in many bacteria

telomeres chromosome tips

centromeres regions of the chromosome linking chromatids

microtubules protein strands within the cell, parts of the cytoskeleton



Fluorescence *in situ* hybridization (FISH) visualization of one of the artificial human chromosomes successfully created by investigators at the Case Western Reserve University School of Medicine and Athersys, Inc. The arrow highlights the synthetic microchromosome, a miniature version of the surrounding native human chromosomes. The chromosome material appears dark blue, and the centromeres white.

cerevisiae), telomeres, centromeres, and origins of replication have all been defined using genetics and have been cloned. When assembled they can be grown as a small chromosome in bacteria and form a **vector** capable of incorporating up to a million bases of other DNA as a chromosome in yeast (a YAC).

vector carrier

This technology has been used to investigate the properties of yeast chromosomes but has been most extensively used in the early phases of genome mapping projects. By cloning complete representations of the human genome into large YACs, the order of these YACs could be deduced by a number of methods and overlapping ones assembled conceptually into a representation of regions of the human genome. These are useful for finding genes from information about the inheritance of genetic diseases. They have also been useful for testing the function of genes in mice. Because of the size of YACs, they frequently contain all of the DNA needed to control the expression of genes with the correct developmental and tissue specificity. When injected into developing mouse eggs, they can fully correct mutations. More recently YACs have been largely supplanted by BACs, because the latter are easier to manipulate and prepare in the laboratory.



Mammalian Artificial Chromosomes

Mammalian artificial chromosomes (MACs) are conceptually similar to YACs, but instead of yeast sequences they contain mammalian or human ones. In this case the telomeric sequences are multimers (multiple copies) of the sequence TTAGGG, and the commonly used centromeric sequence is composed of another repeated DNA sequence found at the natural centromeres of human chromosomes and called alphoid DNA.

Because the alphoid DNA is needed in units of many kilobases, these MAC DNAs are grown as YACs or, more recently, as BACs. When added to suitable cell lines, these MAC DNAs form chromosomes that mimic those in the cell, with accurate segregation and the normal complement of proteins at telomeres and centromeres. Their primary use is not in genome mapping but as vectors for delivery of large fragments of DNA to mammalian cells and to whole animals for expression of large genes or sets of genes. They are still in development, and although gene expression has been demonstrated they have not been used in a practical application. **SEE ALSO** CHROMOSOME, EUKARYOTIC; HUMAN GENOME PROJECT; MAPPING; TELOMERE.

Howard Cooke

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Classical Hybrid Genetics

Common garden peas (*Pisum sativum*) are wonderful when eaten raw. Gregor Mendel (1822–1884) no doubt ate his share. Today he is recognized for using peas to establish the science of genetics.

Mendel investigated hereditary patterns of hybrids. Hybrids are offspring from two organisms that are of different breeds, varieties, or species. Hybrids create new **cultivars**, from new apple varieties to tangelos to hybrid corn. Some mammals produce hybrids; a mule is the progeny of a horse and a donkey.

Mendel was interested in new flower varieties and absorbed by what hybrids reveal about inheritance. Nineteenth-century scientists wanted to know how organisms created a vast diversity of forms while faithfully maintaining distinct sets of characteristics. Constancy was a known quality of life. Then, as now, people had no trouble recognizing the difference between a housefly and a bee. But what prevented people from suddenly sprouting flowers or losing their human features? Genes, DNA, meiosis, and chromosomes were all unknown. Hybrid genetics was a means to discover answers to fundamental biological questions.

Mendel's aim was to discover the mathematical rules behind the reappearing patterns he saw in hybrids. After testing many varieties of peas, he decided to study seven specific traits: shape of the ripe seeds, seed color,

cultivars plant varieties resulting from selective breeding

seed coat color, form of the ripe pods, color of the unripe pods, position of the flowers, and length of the stems.

Initially, Mendel determined the inheritance patterns for one trait at a time, ignoring other traits. He crossed two varieties that differed sharply in a specific trait. For instance, for the trait of seed shape, he used one variety whose seeds were wrinkled and another whose seeds were smooth. The seed-shape trait was represented by two specific inherited features, smooth and wrinkled, that were **alleles**.

Mendel also crossed plant varieties, each with specific combinations of alleles, in order to follow the fates of two or even three traits at the same time. For instance, he crossed a tall plant with wrinkled, green seeds and a short plant with round, yellow seeds. (The underlined words represent alleles for three traits: height, seed shape, and seed color.) To cross two varieties, Mendel first had to make sure the flowers did not pollinate themselves. To do that, he cut off their anthers, their pollen-bearing parts. Then he used the anthers from one variety to pollinate another.

Mendel was the first person to follow specific alleles (which he called “factors”) as they were passed from parental varieties through several generations of offspring. He selected plant varieties that were true-breeding: Each generation of plants looked like the antecedent generations, with regard to the studied traits. Mendel used artificial pollination (described above) to perform an initial outcross, a mating between individuals differing in their alleles for at least one trait. The outcross created hybrids, or, more precisely, **heterozygotes** for the chosen traits. Next, Mendel let the heterozygotes self-pollinate, or intercross (i.e., the mating of inbred animals or self-pollinating plants), that are heterozygous for one or more traits. He then let two subsequent generations of **progeny** self-pollinate.

Mendel discovered that hybrids (heterozygotes) resembled only one parental variety, despite clearly identified input from both parents. Regarding the stem-length trait for example, one parental variety had tall stems, whereas the other variety had very short (dwarf) stems, yet all their hybrid offspring had only tall stems. Mendel named the displayed feature “dominant,” because the hybrid’s appearance of a tall stem (**phenotype**) reflected input from only one parent. He identified the dominant factor or allele for each of the seven traits (seed shape, seed color, length of stem, etc.). The other hidden feature (named “recessive” by Mendel) resurfaced in the next generation, when both hybrid (heterozygous) parents donated their hidden (recessive) allele to a descendent. For example, dwarf stems appeared again in progeny, instead of the conspicuous tall stems seen in the heterozygous parents. A descendent with dwarf stems had a “homozygous recessive” **genotype** for this trait. An individual’s genotype described the two alleles received for a trait, whether hidden or displayed. If a plant had a dominant phenotype for one or several traits, but the genotype was unknown, a method called a test cross was used to reveal the presence of a recessive allele for each of the traits in question. The dominant phenotype to be tested might carry one dominant and one recessive allele, or two dominant alleles (see Mendel’s laws, below), depending on what its parents had donated. The dominant phenotype was crossed with a plant known to be homozygous recessive for a particular trait. If half the offspring showed the dominant phenotype and the other half the recessive

alleles particular forms of genes

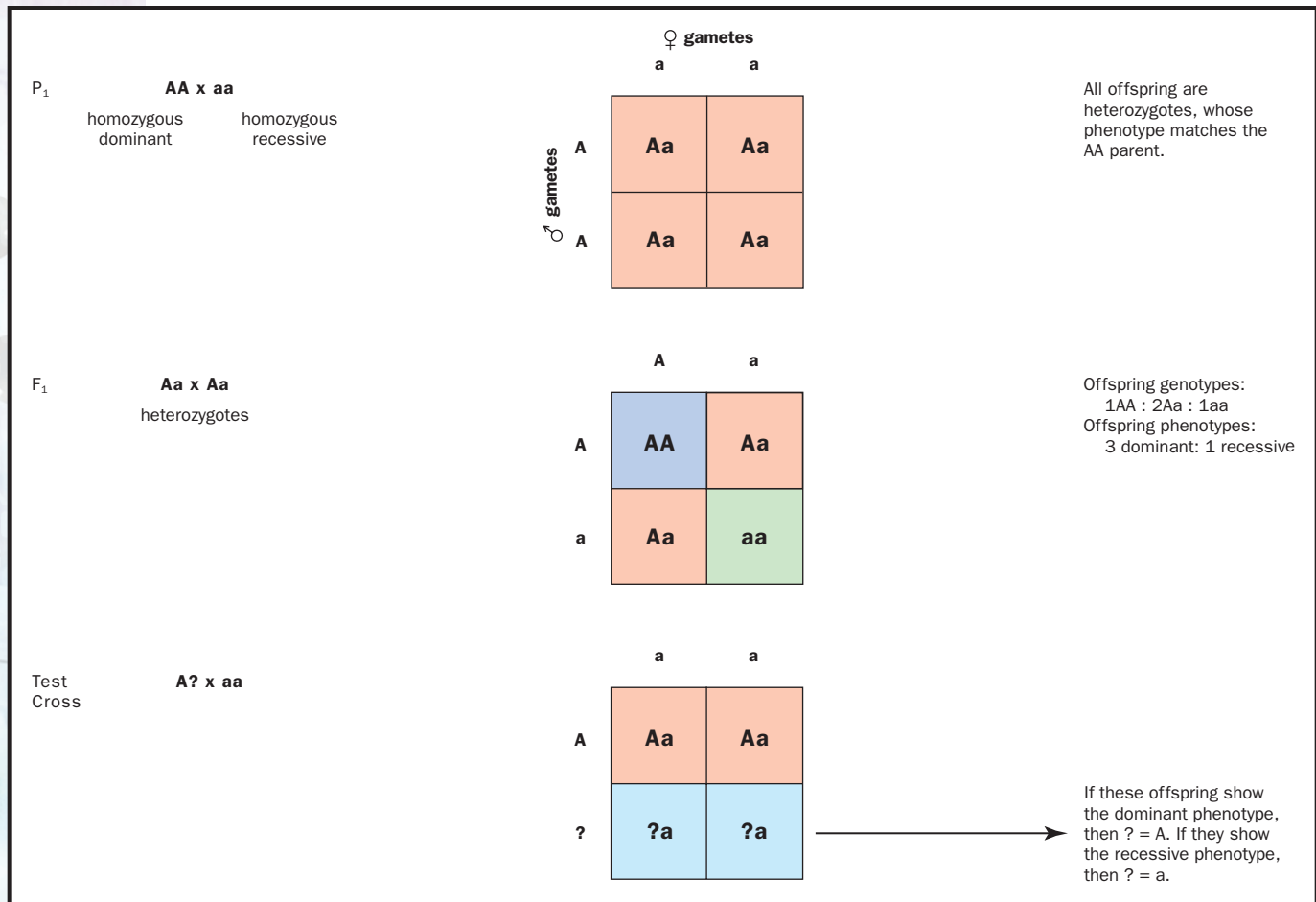
heterozygotes individuals whose genetic information contains two different forms (alleles) of a particular gene

progeny offspring

phenotype observable characteristics of an organism

genotype set of genes present





Punnett squares are used to track the inheritance of traits. The upper- and lowercase letters represent dominant and recessive alleles, respectively. P₁ is the parental generation, and F₁ is the first filial, or offspring, generation. A test cross is performed to determine the genotype of an organism showing the dominant trait, but whose genotype is unknown.

form, then the parent with the dominant phenotype had to carry one recessive and one dominant allele for that trait. In contrast, if all its offspring showed only the dominant form of the trait, the plant with the dominant phenotype must have contained only dominant alleles.

By categorizing and counting offspring of several generations of plants, Mendel discovered two laws of inheritance, described as segregation and independent assortment. Segregation meant that a gamete or reproductive cell received only one allele out of a choice of two alleles carried for each trait (gene) within a parent cell. The gamete had an equal chance of receiving either allele. Independent assortment indicated that traits entered into a gamete independently of each other. This is only true for gene or trait collections located on long strings of hereditary material (chromosomes) that are now termed “nonhomologous” (not alike). Nonhomologous chromosomes carry unique collections of specific genes or traits, whereas two homologous chromosomes carry the same collection of genes, but may carry two distinct alleles for a specific trait or gene.

Armed with his laws, Mendel was able to predict the frequency at which various alleles for several traits would co-appear in descendants of hybrids of the common garden pea. Mendel’s principles were not appreciated for

Back crosses, which are based on Mendel's principles, are used to develop commercially useful plant or animal varieties. In a back cross, a heterozygous plant and its offspring are crossed repeatedly to one of their parents to develop a line of plants that mostly resemble one parent but that have an allele of interest from the other. For instance, a rare flower color is integrated into a line of nematode-resistant stock of roses, joining beauty and disease-resistance. SEE ALSO GENETICS; INHERITANCE PATTERNS; MENDEL, GREGOR; MENDELIAN GENETICS; PROBABILITY.

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Clinical Geneticist

Medical genetics is the application of genetics to the study of human health and diseases. As a profession, medical genetics is usually a mixture of both clinical services and research. Worldwide, services can include diagnosis, counseling, and management of birth defects and genetic disorders. How medical genetics is actually practiced depends on several factors, including the expertise and training of the professionals involved, the expertise available within any given medical facility, and the structure of the practice of medicine within a given society.

In the United States, the practice of medical genetics includes two different career tracks, both requiring certification by the American Board of Medical Genetics (ABMG): the medical geneticist and the clinical geneticist. A medical geneticist holds a Ph.D. and is certified in medical genetics. Typically a medical geneticist is a highly trained research laboratory professional who can additionally take on the role of consultant to physicians. A clinical geneticist is a physician (either a doctor of osteopathy or a medical doctor) involved in all parts of clinical practice related to genetic disorders. Working closely with patients, clinical geneticists identify, diagnose, determine the prognosis of, develop predictive tests for, treat, and manage genetic diseases. They can also be active in conducting research on genetic disorders and studying theoretical genetics, and they usually help to administer and set policies for the clinical genetics profession and for medical centers in general.

Clinical geneticists also can be involved in the bioethical debates and policy-making issues concerning how genetic information is gathered, who has access to it, and how that access should be regulated. This role is becoming increasingly important as society struggles to deal with the tremendous explosion of genetic information arising as a consequence of the Human

Genome Project. Administrative roles for clinical geneticists can include formulating plans and procedures for clinical genetic services, scheduling the use of medical genetics facilities, and teaching interns and residents the methods and procedures involved in the diagnosis and management of genetic disorders.

Very few clinical geneticists develop a private practice. Instead, they typically work in a team environment within regional medical centers alongside scientists, medical geneticists, genetic counselors, and other academics. Most hospitals specializing in pediatric care will also have clinical geneticists on their staff. Some large national clinics have entire departments devoted to the practice of medical and clinical genetics.

Physicians are attracted to the practice of clinical genetics for a variety of reasons. Many enjoy understanding the evolving human gene map, the rapid technological advances in the field, and the opportunity to perform laboratory research as well as practice medicine. They enjoy the challenge of applying the advances in the molecular basis of disease to the care of patients. As a group, clinical geneticists derive satisfaction by remaining close to the “cutting edge” of new discoveries in genetic diseases, which challenges them to remain current while constantly using their knowledge and skills to provide innovative and effective medical services.

Many are attracted to the profession because it allows them to develop long-term relationships with patients and their families. Others find that the narrow focus of clinical genetics is more to their liking than the broader disciplines of internal medicine or pediatrics. Within clinical genetics, physicians can develop their own disease specialty if they choose, which for some provides a more rewarding work environment, giving them the opportunity to make an impact on both research and the lives of patients whose diseases may be rare and often poorly understood by other medical practitioners. Clinical geneticists enjoy complex problem-solving, taking care of people, and paying attention to details that others may miss. They are good listeners.

Students interested in clinical genetics as a profession should become familiar with mathematics, chemistry, biology, and some physics, while still in high school. Courses aimed at developing communication and writing skills are also valuable for students preparing for a career in clinical genetics. Because the practice of clinical genetics requires a medical degree, students must first receive a bachelor's degree, enrolling in courses that meet medical school admission requirements.

After obtaining a medical degree, clinical geneticists typically complete three to five years of residency in medical disciplines approved by the Accreditation Council for Graduate Medical Education (ACGME), followed by a two- to three-year fellowship that is approved by the ABMG, in clinical genetics itself. Certification can be in clinical genetics or in more focused subspecialties, including clinical cytogenetics, clinical biochemical genetics, clinical molecular genetics, and molecular genetic pathology. Certification requires the successful passage of a national examination that is given at regularly scheduled intervals. To maintain certification, clinical geneticists must fulfill continuing education requirements throughout the duration of their career.

Both the ACGME and ABMG maintain a list of approved programs that lead to certification as a clinical geneticist. Approved residency programs include clinical and academic components. Approved programs expose the resident to a patient population large enough to develop an understanding of the wide variety of medical genetic problems. They enable direct involvement in genetic research laboratories, where students learn to critically interpret laboratory data. They include graduate-level course work in basic, human, and medical genetics, as well as clinical teaching conferences, and they foster the development of the communication skills necessary to interact and sustain a long-term therapeutic relationship with patients and their families.

One need not necessarily decide upon the profession of clinical genetics prior to entering medical school or even upon receiving a medical degree or completing a full residency. Traditionally, the ABMG has accepted physicians into clinical genetics programs who come from approved residency programs in pediatrics, obstetrics-gynecology, and internal medicine. This is because the profession of clinical genetics originated from the treatment of inherited diseases that are initially observed in newborn infants and children. It is still within these disciplines that many physicians develop a secondary interest in clinical genetics, during residency or even later in their medical career. However, the field of medical genetics is rapidly changing, due to the recognition that many genetic disorders result in symptoms that are delayed until adulthood. As a consequence, clinical geneticists are now beginning to provide their services to adults as well.

The Human Genome Project has also brought the realization that clinical genetics involves more than single-gene and single-chromosome conditions. Genetic medicine is becoming applicable to many kinds of complex diseases and disorders such as cancer, heart disease, and asthma, to name a few well-known conditions.

The 1,100 certified clinical geneticists registered in the United States in the year 2000 were too few to keep up with an ever-increasing demand for their services. Students interested in pursuing clinical genetic careers can expect that the number of subspecialties will continue to grow, that both the ACGME and ABMG will continue to strive to provide new certification programs and career tracks, and that the concept of clinical genetics may become an integral component of “well” medical health care. SEE ALSO DISEASE, GENETICS OF; GENETIC COUNSELOR; GENETIC TESTING; GENETIC TESTING: ETHICAL ISSUES; PRENATAL DIAGNOSIS.

Diane C. Rein

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Cloning Genes

Gene cloning, or molecular cloning, has several different meanings to a molecular biologist. A clone is an exact copy, or replica, of something. In the literal sense, cloning a gene means to make many exact copies of a segment of a DNA molecule that encodes a gene. This is in marked contrast to cloning an entire organism—regenerating a genetically identical copy of the organism—which is technically much more difficult (with animals) and can involve ethical ramifications not associated with gene cloning. Molecular biologists exploit the replicative ability of cultured cells to clone genes.

Purposes of Gene Cloning

To study genes in the laboratory, it is necessary to have many copies on hand to use as samples for different experiments. Such experiments include Southern or Northern blots, in which genes labeled with radioactive or fluorescent chemicals are used as probes for detecting specific genes that may be present in complex mixtures of DNA.

Cloned genes also make it easier to study the proteins they encode. Because the genetic code of bacteria is identical to that of **eukaryotes**, a cloned animal or plant gene that has been introduced into a bacterium can often direct the bacterium to produce its protein product, which can then be purified and used for biochemical experimentation. Cloned genes can also be used for DNA sequencing, which is the determination of the precise order of all the base pairs in the gene. All of these applications require many copies of the DNA molecule that is being studied.

Gene cloning also enables scientists to manipulate and study genes in isolation from the organism they came from. This allows researchers to conduct many experiments that would be impossible without cloned genes. For research on humans, this is clearly a major advantage, as direct experimentation on humans has many technical, financial, and ethical limitations.

Cloning Techniques

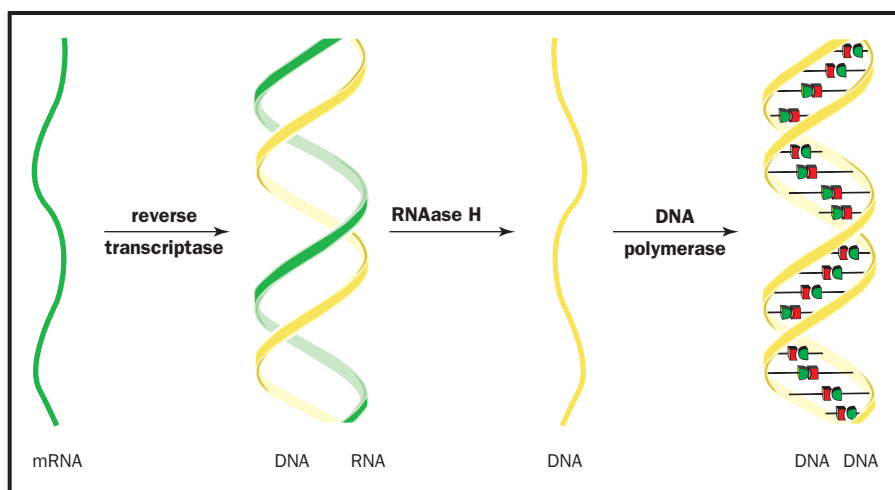
Cloning genes is now a technically straightforward process. Usually, cloning uses recombinant DNA techniques, which were developed in the early 1970s by Paul Berg, of Stanford University, and, independently, by Stanley Cohen and Herbert Boyer, of Stanford and the University of California. These researchers devised methods for excising genes from DNA at precise positions, using **restriction enzymes** and then using the enzyme known as DNA ligase to splice the resulting gene-containing fragment into a plasmid **vector**.

Plasmids are small, circular DNA molecules that occur naturally in many species of bacteria. The plasmids naturally replicate and are passed on to future generations of bacterial cells. To replicate, all plasmids must contain a sequence, called an origin of replication, which directs the bacterial DNA

eukaryotes organisms with cells possessing a nucleus

restriction enzymes enzymes that cut DNA at a particular sequence

vector carrier



cDNA is double-stranded DNA that is synthesized from single-stranded messenger RNA. cDNA has all the coding and regulatory regions of the original gene, but no introns.

polymerase to replicate the DNA molecule. In addition, recombinant plasmids contain one or more selectable markers. A selectable marker is a gene that confers on the bacterium harboring the plasmid the ability to survive under conditions in which bacteria lacking the plasmid would otherwise die. Usually, such genes encode enzymes that enable the bacterium to live and grow despite the presence of an antibiotic drug.

The recombinant plasmid is then introduced into a host cell, such as an *Escherichia coli* bacterium, by a process called transformation, and the cell is allowed to multiply and form a large population of cells. Each of these cells harbors many identical copies of the recombinant plasmid. The cells are then cultured in growth media containing the antibiotic to which the plasmid confers resistance. This ensures that only cells containing the recombinant plasmid will survive and replicate. A researcher then harvests the cells and can extract and purify many copies of the plasmid.

Another method to produce many copies of a DNA molecule, which is even simpler than traditional recombinant cloning methods, is the polymerase chain reaction (PCR). PCR amplifies the DNA in a reaction tube without the need for a plasmid to be grown in bacteria.

Importance for Medicine and Industry

The ability to clone a gene is not only valuable for conducting biological research. Many important pharmaceutical drugs and industrial enzymes are produced from cloned genes. For example, insulin, clotting factors, human growth hormone, cytokines (cell growth stimulants), and several anticancer drugs in use are produced from cloned genes.

Before the advent of gene cloning, these proteins had to be purified from their natural tissue sources, a difficult, expensive, and inefficient process. Using recombinant methods, biomedical companies can prepare these important proteins more easily and inexpensively than they previously could. In addition, in many cases the product that is produced is more effective and more highly purified. For example, before the hormone insulin, which many diabetes patients must inject, became available as a recombinant human protein, it was purified from pig and cow pancreases. However, pig and cow insulin has a slightly different amino acid sequence than the

polymerase enzyme complex that synthesizes DNA from individual nucleotides

Escherichia coli common bacterium of the human gut, used in research as a model organism

human hormone. This sometimes led to immune reactions in patients. The recombinant human version of the hormone is identical to the natural human version, so it causes no immune reaction.

Gene cloning is also used to produce many of the molecular tools used to study genes. Even restriction enzymes, DNA ligase, DNA polymerases, and many of the other enzymes used for recombinant DNA methods are themselves, in most cases, produced from cloned genes, as are enzymes used in many other industrial processes.

Genomic Versus cDNA Clones

A gene can take varying forms, and so can gene clones. The protein-coding regions of most eukaryotic genes are interrupted by noncoding sequences called introns, which are ultimately excluded from the mature messenger RNA (mRNA) after the gene is transcribed. In addition to the protein-coding sequences, all genes contain “upstream” and “downstream” regulatory sequences that control when, in which tissues, and under what circumstances the gene is transcribed. A clone containing the entire region of a gene as it exists on the chromosome, including introns and nontranscribed regulatory sequences, is called a genomic clone because it is derived directly from genomic, or chromosomal, DNA.

It is also possible to clone a gene directly from its messenger RNA transcript, from which all introns have been removed. This type of clone, called a complementary DNA or cDNA clone, includes only the protein-coding sequences and upstream and downstream sequences that do not code for amino acids but that may control how the mRNA transcript gets translated to protein.

To prepare cDNA a researcher starts with mRNA and then makes a complementary single-stranded DNA copy using the enzyme reverse transcriptase. Reverse transcriptase is a DNA polymerase that synthesizes DNA based on an RNA template that is produced by retroviruses. After the mRNA strand is digested away by another enzyme, called RNase H, DNA polymerase can synthesize a second DNA strand by using the newly made first strand cDNA as a template.

Because cDNAs lack introns, the protein-coding region in a cDNA molecule is contained in a single uninterrupted sequence, called an open reading frame, or ORF. This makes cDNA clones extremely useful for predicting the amino acid sequence of the protein that a gene encodes. It also makes it possible to direct protein synthesis from a eukaryotic cDNA clone in a bacterium, which cannot splice introns. With introns still present in a cloned gene, the bacteria will misinterpret the intron sequences as protein-encoding sequences. The resulting incorrect messenger RNA will encode a protein with an incorrect amino acid.

“Gene Cloning” Usually Means “Gene Identification”

When researchers report in a scientific journal that they have “cloned a gene” they are not referring to the rather mundane process of amplifying copies of a DNA molecule. What they are really talking about is the molecular identification of a previously unknown gene, and determination of its precise position on a chromosome. There are many different methods that

can be used to identify a gene. Two of the most common approaches are discussed below.

A gene can be defined in several ways. In fact, the concept of the gene is undergoing a re-evaluation as scientists are analyzing the complete genomes of more and more organisms and finding that many sequences encode more than one protein product. Gregor Mendel identified genes—for example, he identified the factor that made peas either yellow or green—long before he or anyone else knew that genes were encoded on segments of the DNA that made up chromosomes. Studying genetics in the fruit fly, *Drosophila melanogaster*, Morgan and Sturtevant demonstrated that genes are entities that reside at measurable locations, or loci, on chromosomes, although they did not yet understand the biochemical nature of genes.

Modern geneticists often use the same methods as Mendel and Morgan to identify genes by physical traits, or phenotypes, that mutations in them can cause in an organism. But today we can go even further. Using a broad range of molecular biology techniques, including gene cloning, researchers can now determine the precise DNA coding sequence that corresponds to a particular **phenotype**. This capability is tremendously powerful, because discovering the gene responsible for a trait can help humankind understand the cellular and biochemical processes underlying the trait. For example, geneticists have learned a great deal about the basis of cancer by identifying genes that, when mutated, contribute to cancer. By studying these genes, researchers now know that many of them control when cells divide (e.g., proto-oncogenes and tumor suppressor genes) or when they die (e.g., the apoptosis genes). Under some circumstances, when such genes are damaged by mutation, cells divide when they shouldn't, or don't die when they should, leading to cancer.

phenotype observable characteristics of an organism

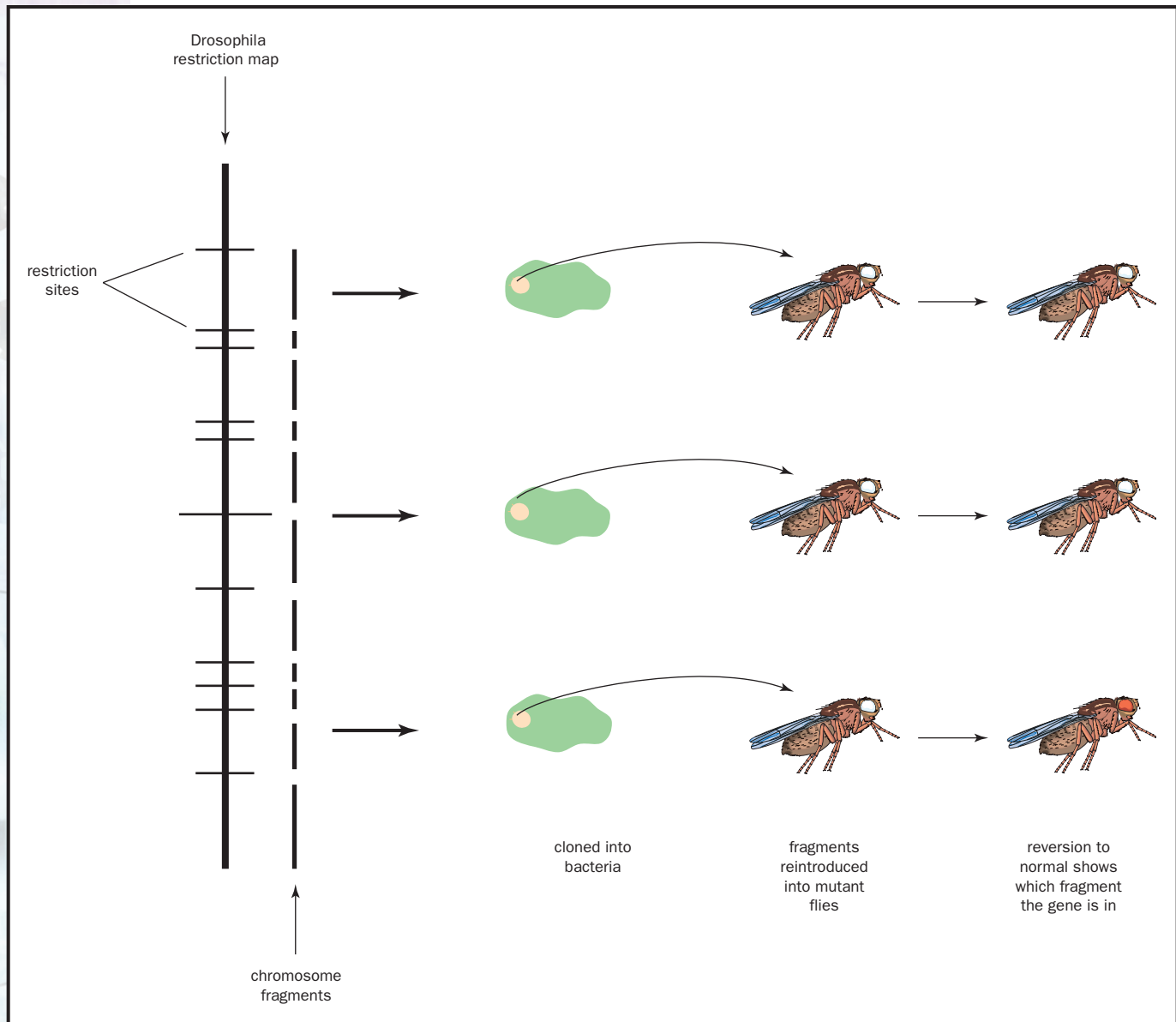
Positional Cloning

Positional cloning starts with the classical methods developed at the turn of the twentieth century by Thomas Hunt Morgan, Alfred Sturtevant, and their colleagues, of genetically mapping a particular phenotype to a region of a chromosome. A detailed discussion of genetic mapping is beyond the scope of this section, but, in general, it is based on conducting genetic crosses between individuals with two different mutant traits and analyzing how often the traits occur together in the progeny of subsequent generations.

Genetic mapping provides a general idea of where a gene is located on a particular chromosome, but it does not identify the precise DNA sequence that encodes the gene. The next step is to locate the gene on what is called the physical map of the chromosome. A physical map is a high-resolution map of all the DNA sequences that make up a chromosome. One type of physical map is a restriction map, which depicts the order of DNA fragments produced when a large DNA molecule is cut with restriction endonucleases (restriction enzymes).

Restriction maps have been made for the complete genomes of several model genetic organisms, such as the fruit fly (*Drosophila melanogaster*), and the roundworm, (*Caenorhabditis elegans*). For these organisms, individual large DNA fragments—on the order of forty to one hundred thousand base pairs from the whole genome—have been cloned in bacterial plasmid vectors to make a “library” of the genome. Each fragment is mapped to a known

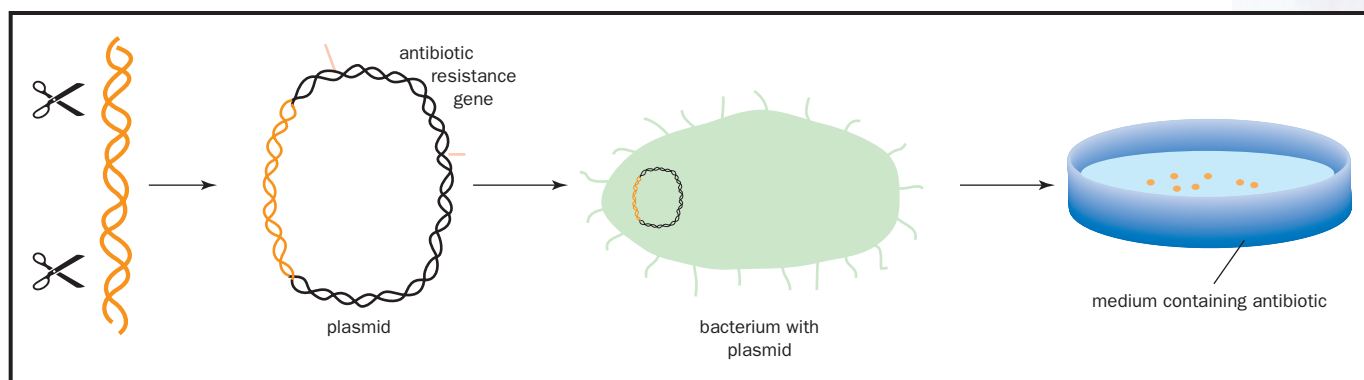




The location of a gene can be found by cutting the DNA with restriction enzymes, inserting the fragments into bacteria, and then reintroducing them individually into mutant flies.

position, but the identify of the gene or genes it contains is originally unknown. To identify the genes, a cloned fragment is introduced into a mutant fly or roundworm.

To pinpoint the location of a particular gene, a researcher can introduce one or several of the plasmid clones from the physical map that are in the general vicinity of the region on the genetic map where the gene is thought to lie into a mutant that is defective in the gene of interest. If the introduced DNA corrects the mutant's defect, that DNA probably contains a normal copy of the defective gene. But these large clones usually contain several genes. By further "trimming" the DNA into smaller subfragments and testing the ability of each subfragment to rescue mutants, the researcher can eventually home in on the gene. As further confirmation that this gene is the cause of the mutant phenotype, the researcher can isolate the corresponding gene from the mutant and determine its DNA sequence to see if



it contains a mutation (a DNA sequence alteration) relative to the normal gene sequence.

Expression Cloning

In some cases, a researcher becomes interested in studying a gene not because mutations in it cause an interesting phenotype but because the protein it encodes has interesting properties. A prominent example is beta-amyloid protein, which accumulates in the brains of Alzheimer's disease patients.

Expression cloning is a method of isolating a gene by looking for the protein it encodes. If the protein of interest is an enzyme, it can be found by testing for its biochemical activity. A very common method for identifying a particular protein is by using antibodies, or immunoglobulins, that bind specifically to that protein. Expression cloning usually uses a cDNA library, in which protein-coding sequences are uninterrupted by introns. Each cDNA is inserted into an "expression vector," which contains all the necessary signals for the DNA to be transcribed into mRNA. The mRNA can then be translated into protein. Thus the host cell harboring the clone will produce the gene's protein product, and the protein can then be detected by biochemical or immunologic methods. Once the cell making the protein is found, the cDNA can be re-isolated and the gene sequenced by standard means.

Gene cloning techniques continue to advance rapidly, aided by the Human Genome Project and bioinformatics. It is likely that positional cloning will take on a secondary role, and that bioinformatics and proteomics methods will begin to contribute more, as more progress in these fields is made. SEE ALSO BIOINFORMATICS; BLOTTING; CHROMOSOMES, ARTIFICIAL; CLONING ORGANISMS; CLONING: ETHICAL ISSUES; DNA LIBRARIES; GENE; GENE DISCOVERY; HUMAN GENOME PROJECT; LINKAGE AND RECOMBINATION; MARKER SYSTEMS; MORGAN, THOMAS HUNT; PLASMID; POLYMERASE CHAIN REACTION; RECOMBINANT DNA; RESTRICTION ENZYMES; REVERSE TRANSCRIPTASE; RNA PROCESSING; SEQUENCING DNA; TRANSFORMATION.

Paul J. Mublrud

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Restrictive enzymes (scissors) cut a gene out of its normal chromosomal position. Other enzymes insert it into a plasmid, which is then introduced into a bacterium. Only those bacteria that took up the plasmid survive on the growth medium. These bacteria can then be grown in bulk to produce many gene copies.

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Cloning: Ethical Issues

Cloning is the creation of an individual that is a genetic replica of another individual. The process transfers a nucleus from a somatic nonreproductive cell into an “enucleated” fertilized egg, one that has had its own nucleus destroyed or removed. The genes in the transferred nucleus then direct the development of a complete organism from the altered fertilized egg. Two individuals who are clones have identical genes in their cell nuclei, but differ in characteristics that are acquired in other ways.

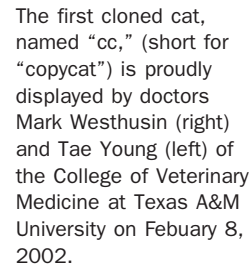
Cloning in Context

Cloning is a natural phenomenon in species as diverse as armadillos, poplar trees, aphids, and bacteria. Identical twins are clones. Biologists have been cloning some organisms, such as carrots, for decades. Attempts to clone animals have been far less successful. They began long before the February 1997 announcement of the birth of Dolly, a sheep cloned from a mammary gland cell nucleus of a six-year-old sheep.

Oxford University developmental biologist John Gurdon cloned frogs in the 1960s, but in a limited way. He showed that a nucleus from a tadpole’s intestinal lining cell could be transferred to an enucleated fertilized egg and support development to adulthood, and that a nucleus from an adult cell could support development as far as the tadpole stage. However, he was unable to coax a nucleus from an adult amphibian’s cell to support development all the way to adulthood. In the 1980s several companies tried to commercialize cloning of livestock from nuclei taken from embryos or fetuses. The efforts failed because the cloned animals were nearly always very unhealthy newborns and did not survive for long. Currently, livestock cloning is limited to research, although some companies offer tissue preservation services in anticipation of future advances in commercial livestock cloning. There is no reason to believe that human clones would fare any better in terms of health or survivability than most cloned animals do.

The Cloning Ban

Ethical concerns about whether an action is “right” or “wrong” are often clouded by subjectivity, emotion, and perspective. Cloning members of an endangered species, for example, is generally regarded as a positive application of the technology, whereas attempting to clone an extinct woolly mammoth from preserved tissue elicits more negative responses, including that this interferes with nature. A project at Texas A&M University, funded by a dog lover wishing to clone a beloved deceased pet, announced the first successful cloning of a domestic animal, a cat, in February 2002. Cloning pets when strays crowd shelters might be seen as unethical. A different set of ethical issues emerges when considering the cloning of humans, which a few scientists and physicians have proposed doing outside of the United States.



According to the bill's ban on producing or selling "any embryo produced by human cloning," scientists caught in the act could expect a fine of up to \$1 million or ten years in prison. Proposals to exempt therapeutic

cloning were defeated. The criminalization of basic research is unprecedented: Before 2001, bans on using embryonic stem cells applied only to federally funded research, and work using a small number of previously existing stem cell lines was permitted. Since the 2001 ruling, some researchers have moved to nations that permit them to derive new embryonic stem cell lines. Stem cells that are normal parts of adult bodies are being investigated as alternative sources of replacement tissues.

Cloning Misconceptions

The premise that a clone is an exact duplicate of another individual is flawed, and so if the intent of cloning is to create such a copy, it simply will not work. For example, the tips of chromosomes, called **telomeres**, shorten with each cell division. A clone's telomeres are as short as those from the donor nucleus, which means that they are "older" even at the start of the clone's existence. DNA in the donor nucleus has also had time to mutate, that is to say, it has had time to undergo modification from its original sequence, thus distinguishing it genetically from other cells of the donor. A mutation that would have a negligible or delayed effect in one cell of a many-celled organism, such as a cancer-causing mutation, might be devastating if an entire organism develops under the direction of that nucleus. Finally, the clone's **mitochondria**, the cell organelles that house the reactions of metabolism and contain some genes, are those of the recipient cell, not the donor, because they reside in the cytoplasm of the egg. Mitochondrial genes, therefore, are different in the clone than they are in the nucleus donor. The consequences of nuclear and mitochondrial genes from different individuals present in the same cell are not known, but there may be incompatibilities.

Perhaps the most compelling reason why a clone is not really a duplicate is that the environment affects gene expression. Cloned calves have different color patterns, because when the animals were embryos, the cells that were destined to produce pigment moved in different ways in each calf. For humans, consider identical twins. Nutrition, stress, exposure to infectious diseases, and other environmental factors greatly influence our characteristics. For these reasons, cloning a deceased child, the application that most would-be cloners give for pursuing the technology, would likely lead to disappointment.

Bioethical concerns over cloning may be moot, because the procedure is extremely difficult to do. Dolly was one of 277 attempts; Cumulina, the first cloned mouse, was among 15 liveborn mice from 942 tries. Cloning so often fails, researchers think, because it is not a natural way to start the development of an animal. That is, the DNA in a somatic cell nucleus is not in the same state as the DNA in a fertilized **ovum**. The donor DNA in cloning does not pass through an organism's germ line, the normal developmental route to sperm or egg, where gene activities are regulated as a new organism develops.

Ethical objections to human cloning are more philosophical than they are practical. The very idea of cloning assumes that our individuality can be understood so well that we can duplicate it. If human cloning ever became a reality, that this is not true would become evident. After all, we are more than a mere collection of genes. SEE ALSO BIOTECHNOLOGY: ETHICAL ISSUES;

telomeres chromosome tips

mitochondria energy-producing cell organelle

ovum egg

CLONING GENES; CLONING ORGANISMS; MITOCHONDRIAL GENOME; STEM CELLS; TELOMERE.

Ricki Lewis

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Cloning Organisms

There are two distinct types of cloning: molecular and organismal. Molecular cloning is the removal of a stretch of DNA, usually a gene, from an organism, and its insertion into another piece of DNA, such as a **plasmid**, to form a substance called recombinant DNA. This recombinant DNA may then be expressed in, or simply carried passively by, another organism, such as bacteria. Organismal cloning, the subject of this entry, is the production of genetically identical organisms and, as such, can be used to produce genetically identical copies of livestock or may be used to produce new members of endangered or even extinct species. It may be especially cost-effective to clone animals that produce therapeutic proteins such as blood clotting factors, thus combining both types of cloning. Cloning is controversial, however, because our understanding of the procedures needed to clone mammals may be applied to human cloning, which gives rise to profound ethical issues.

plasmid a small ring of DNA found in many bacteria

The History of Cloning

Cloning has a long history. Animals that reproduce sexually produce clones whenever identical twins are born. These twins are genetically indistinguishable, and are formed when a fertilized egg separates at a very early stage of development. Clones are also the natural product of asexual reproduction, although in this case perfect clones cannot be maintained through an infinite number of generations, because spontaneous mutations can and do occur. Lastly, clones can be produced by regeneration in both plants and animals. For example, plant cuttings will regenerate roots and, ultimately, an entire "new" plant, and some invertebrates, such as planaria, can regenerate two identical animals if the adult is cut in half. In these forms, cloning has been with us for a very long time.

Since the mid-1960s, scientists have been able to culture plant cells, that is, grow cells from plants such as tobacco and carrots in a petri dish, to get thousands of genetically identical cells. From such cultured cells an unlimited quantity of cloned plants can then be grown. These cultured cells can be modified to contain recombinant, or cloned, DNA as well.

Cloning Amphibians

The first cloning of a vertebrate by nuclear transfer was reported by John Gurdon of the University of Cambridge in the 1950s. In nuclear